सेंट्रल ट्रांसमिशन यूटिलिटी ऑफ इंडिया लिमिटेड
(पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड के स्वामित्व में)
(भारत सरकार का उदयम)

CENTRAL TRANSMISSION UTILITY OF INDIA LTD.
(A wholly owned subsidiary of Power Grid Corporation of India Limited)
(A Government of India Enterprise)

Ref: CTU/N/00/CMETS_NR/17

Date: 02-05-2023

As per distribution list

Subject: 17th Consultation Meeting for Evolving Transmission Schemes in Northern Region-Minutes of Meeting

Dear Sir/Ma'am,

Please find enclosed the minutes of the 17th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 31st March, 2023 (Friday) through virtual mode.

The minutes are also available at CTU website (www.ctuil.in)

Thanking you,

Yours faithfully,

काशिश 02/05/23

(Kashish Bhambhani)
General Manager (CTU)

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A. Application related matters in Northern Region

1. Proposal for grant of Connectivity to the applications received from Renewable Energy Sources

It was deliberated that following 3 Nos. of Stage-I Connectivity applications were received in the month of Feb'23 and were agreed to be granted as per the details below:

TABLE 1

Sl. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought Date/(MW)	Nature of Applicant	Location requested for Grant of Stage-I Connectivity	Tr. System for Grant of Stage-I Connectivity
1.	0112100048	JSW Renew Energy Five Limited	Jaisalmer distt., Rajasthan	14.02.2023	15.10.2024/ 250	Project based on Standalone Storage Source(s)	Fatehgarh -III PS	JSW Renew Energy Five Limited BESS Project-1 (250 MW) – Fatehgarh-III PS 220 kV cable* – to be implemented by applicant (Suitable to carry minimum 300 MW* at nominal voltage) **Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.**
2.	0112100052	JSW Renew Energy Five Limited	Jaisalmer distt., Rajasthan	14.02.2023	15.10.2024/ 250	Project based on Standalone Storage Source(s)	Fatehgarh -III PS	JSW Renew Energy Five Limited BESS Project-2 (250 MW) – Fatehgarh-III PS 220 kV cable* – to be implemented by applicant (Suitable to carry minimum 300 MW* at nominal voltage) **Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.**

Sl. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought Date/(MW)	Nature of Applicant	Location requested for Grant of Stage-I Connectivity	Tr. System for Grant of Stage-I Connectivity
3.	0112100053	**MRS Buildvision Private Limited	Bikaner distt., Rajasthan	16.02.2023	31.03.2025/ 1000	Renewable Power Park Developer (Solar)	Bikaner -III PS	MRS Buildvision Private Limited Renewable Power Park Project – Bikaner- III PS 400 kV S/c (High Capacity) line- to be implemented by the applicant (Suitable to carry minimum 1000 MW at nominal voltage) **Common Transmission system mentioned at (ii) under ISTS shall also be required for Connectivity.**

**Applicant confirmed during the subject meeting that the DTL shall be 220 kV cable instead of line as its within Fatehgarh-III pooling station which shall be suitable to carry minimum 300 MW at nominal voltage*

*** Grid India Ltd. indicated that considering the park injection quantum (1000 MW), 400 kV D/c line for DTL should be planned. CTU mentioned that as per the CEA Planning Criteria, N-1 condition is not mandatory for RE generating stations DTL. Further, M/s MRS Build vision also stated that they wish to opt for S/c line instead of D/c line, else it will not be economical viable for them.*

(i) Common Transmission system for Connectivity at Fatehgarh-III PS

- Establishment of 400/220 kV pooling station at Fatehgarh-III PS
- Fatehgarh-III PS – Fatehgarh-II PS (Twin HTLS)* 400 kV D/c line or Jaisalmer (RVPN) S/s - Fatehgarh-III PS (Twin HTLS)* 400 kV D/c line.

** with minimum capacity of 2100 MVA on each circuit at nominal voltage*

(ii) Common Transmission system for Connectivity at Bikaner-III PS

- Establishment of Bikaner-III 765/400/220kV PS
- LILO of one Ckt of 400 kV Bikaner (PG)-Bikaner-II (Quad) D/c line at Bikaner-III PS or Bikaner-II – Bikaner-III 400 kV D/c line (Quad)

- It was mentioned that the grant of Stage-I Connectivity shall not create any right in favour of the grantee on ISTS infrastructure including bays. Stage-I Connectivity grantee shall be required to update the development of their generation project and associated transmission infrastructure as per format RCON-I-M (available on CTU website>>Stage-I / Stage-II Status Monitoring Portal) by 30th day of June and 31st December of each year.

- Further Stage-I Connectivity grantee is required to secure the Stage-II Connectivity for physical connectivity with the ISTS grid within 24 months. The Stage-I Connectivity grantees who fail to apply for Stage-II Connectivity within 24 months from grant of Stage-I Connectivity shall cease to be Stage-I Grantee and their Application fees shall be forfeited.

2. Stage-II Connectivity applications

It was deliberated that following 4 Nos. of Stage-II Connectivity applications were received in the month of Feb'23 and were agreed to be granted as per the details below:

TABLE 2

Sl. No.	Application No.	Applicant	Location	Date of Application	Application/ Quantum of Stage-I (MW)	Nature of Applicant	Generation Schedule	Stage-II Connectivity Sought (MW)/date	Quantum won / Land & Auditor Basis	Proposed Location as per St-I Application	Tr. System for Grant of Stage-II Connectivity
1.	0212100037	*Radiant Star Solar Park Private Limited	Barmer distt., Rajasthan	13.02.2023	0112100046 / 300	Renewable Power Park Developer (Solar)	31.12.2024	200/ 31.12.2024 *Revised: 30.09.2025 (Interim)$*	L&A	Fatehgarh -IV (Section-2) PS	➢ Common pooling station of M/s Radiant Star Solar Park Private Limited (200 MW) & M/s Helia Energy Park Private Limited (200 MW) Renewable Power Park Developer (Solar) projects – Fatehgarh -IV (Section-2) PS# 220 kV S/c (high capacity) line on D/c tower – to be implemented by the lead applicant **(Radiant Star Solar Park Pvt Ltd.) (Suitable to carry minimum 400 MW at nominal voltage) ➢ 220 kV line along with 220 kV bay at ISTS PS to be clubbed with Application Nos. 0212100037 & 0212100035 of M/s Radiant Star Solar Park Private Limited & Helia Energy Park Private Limited respectively.
2.	0212100035	*Helia Energy Park Private Limited	Barmer distt., Rajasthan	13.02.2023	0112100043 / 300	Renewable Power Park Developer (Solar)	31.12.2024	200/ 31.12.2024 *Revised: 30.09.2025 (Interim)$*	L&A	Fatehgarh -IV (Section-2) PS	

Sl. No.	Application No.	Applicant	Location	Date of Application	Application/ Quantum of Stage-I (MW)	Nature of Applicant	Generation Schedule	Stage-II Connectivity Sought (MW)/date	Quantum won / Land & Auditor Basis	Proposed Location as per St-I Application	Tr. System for Grant of Stage-II Connectivity
											➢Common 220 kV line bay at Fatehgarh-IV PS (section-2) shall be developed under ISTS. **Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.**
3.	0212100039	##JSW Renew Energy Five Limited	Jaisalmer distt., Rajasthan	14.02.2023	0112100048 / 250	Project based on Standalone Storage Source(s)	15.10.2024	250/ 15.10.2024 *Revised: 30.09.2024 (Interim)$*	(SECI Standalone BESS LoA)	Fatehgarh -III PS	➢JSW Renew Energy Five Limited BESS Project-1 (250 MW) – Fatehgarh-III PS 220 kV cable – to be implemented by applicant (Suitable to carry minimum 300 MW** at nominal voltage) ➢220 kV bay at Fatehgarh-III PS shall be developed under ISTS. **Common Transmission system mentioned at (ii) under ISTS shall also be required for Connectivity.**
4.	0212100040	##JSW Renew Energy Five Limited	Jaisalmer distt., Rajasthan	14.02.2023	0112100052 / 250	Project based on Standalone Storage Source(s)	15.10.2024	250/ 15.10.2024 *Revised: 30.09.2024 (Interim)$*	(SECI Standalone BESS LoA)	Fatehgarh -III PS	➢JSW Renew Energy Five Limited BESS Project-2 (250 MW)** – Fatehgarh-III PS 220 kV cable – to be implemented by applicant (Suitable to carry minimum 300 MW* at nominal voltage) ➢220 kV bay at Fatehgarh-III PS shall be developed under ISTS.

Sl. No.	Application No.	Applicant	Location	Date of Application	Application/ Quantum of Stage-I (MW)	Nature of Applicant	Generation Schedule	Stage-II Connectivity Sought (MW)/date	Quantum won / Land & Auditor Basis	Proposed Location as per St-I Application	Tr. System for Grant of Stage-II Connectivity
											Common Transmission system mentioned at (ii) under ISTS shall also be required for Connectivity.

$ *Final Start date of Connectivity shall be confirmed after award of ISTS works*

**Both the applicants (M/s Radiant Star Solar Park Pvt Ltd. and M/s Helia Energy Park Pvt Ltd.) have mentioned that they have agreed to share common dedicated transmission line & bay at ISTS end for evacuation of 400 MW power from their parks. In this regard, they have earlier submitted lead generator agreement in which Helia Energy Park Pvt. Ltd. shall act as lead generator and undertake all operational and commercial responsibilities on behalf of both the parties. However, during the meeting, applicant informed that now M/s Radiant Star Solar Park Pvt Ltd. shall act as lead generator and undertake all operational and commercial responsibilities on behalf of both the parties. The same was noted and applicant was requested to submit the revised sharing agreement among both the applicants. Applicants noted and subsequently submitted the revised agreement on 18/04/2023. In addition to this, it was also mentioned that no further enhancement shall be granted to M/s Helia & M/s Radiant beyond 400 MW at this 220 kV bay and both the applicants agreed for the same.*

***Applicant confirmed during the subject meeting that the DTL shall be 220 kV cable instead of line as its within Fatehgarh-III pooling station which shall be suitable to carry minimum 300 MW on single cable at nominal voltage.*

It was informed during the meeting that no further margin is available at Fatehgarh-III PS for Connectivity/ LTA in solar peak hours, therefore, applicant may assure that there shall be no power injection from their BESS project during Peak Solar Hours. Further, M/s JSW Energy's clarified on injection aspects as the project being BESS (Storage Project), there will be no injection of power during solar peak hours. They also added that in the RFS document, it was mentioned that BESSD will have to comply with the Charging and Discharging Schedule as intimated by Buying Entity. It is clarified that discharge of BESS shall take place subject to the transmission constraints at the ISTS substation. For example, discharge of power from BESS during peak Solar hours (say, 11:00 AM- 2:00 PM) may be subject to the Grid constraints. CTU indicated that as per the grant of Connectivity & LTA, margins may not be available in peak solar hours, however, it will depend on real time conditions. SECI also confirmed that during Solar Peak hours, BESS is not supposed to inject any power.

Further, NRLDC during the meeting & vide their email dated 06/04/2023 enquired about the operating philosophy of BESS during the day time with its impact on transmission margin and operation during night hours with reactive power perspective as during night hours very high voltages may be there. Regarding this, M/s JSW vide their email dated 12/04/2023 submitted following:

- **Operating philosophy of BESS & impact on transmission margin at ISTS Fatehgarh III PS:**

"They will be building two nos. of standalone Battery Energy Storage Systems (BESS) connected to ISTS Fatehgarh III PS. The total storage capacity of both projects will be 1000 MWh (500 MW x 2 hrs), with the primary objective to make the energy storage facility available to SECI or other buying entities for charging and discharging on an "on demand" basis. They will be developing these projects as per the provisions of SECI's Standard Battery Energy Storage Purchase Agreement (BESPA). Regarding transmission margin at ISTS Fatehgarh III PS, we hereby confirm that we will comply with all the provisions of the BESPA, which mandates that the discharge of BESS shall take place subject to the transmission constraints such as peak solar hours, at the ISTS substation. They will also abide by all application standards set by the CEA, CERC, CTUIL & Grid India."

- **Reactive power perspective:**

"BESS shall be equipped with PCS (Power Conversion System) which is a device that is used to convert power in both directions. The PCS will enable the conversion of power from the grid to DC for charging the batteries, or from the batteries to AC to feed the grid. The PCS is capable of operating in four quadrants, which means that it can both absorb and supply power from and to the grid. The PCS will be designed to comply with the technical requirements for grid connectivity as specified by the Central Electricity Authority (CEA). This includes the ability to export and import both active and reactive power. This means that the PCS will be able to manage the power flow to and from the grid in a controlled manner, ensuring that the grid remains stable and reliable."

(i) Common Transmission system for Connectivity at Fatehgarh-IV PS (under ISTS)

- Establishment 1x500 MVA, 400/220 kV Fatehgarh- IV (Section-2) Pooling Station
- Fatehgarh-IV (Section-2) PS – Bhinmal (PG) 400 kV D/c line (Twin HTLS) *

or

- Establishment of 1x1500 MVA, 765/400 kV & 1x500 MVA, 400/220 kV Fatehgarh- IV (Section-2) Pooling Station
- LILO of both ckts of 765 kV Fatehgarh-III- Beawar D/c line(2nd) at Fatehgarh-IV (Section-2) PS

(ii) Common Transmission system for Connectivity at Fatehgarh-III PS

- Establishment of 400/220 kV pooling station at Fatehgarh-III PS
- Fatehgarh-III PS – Fatehgarh-II PS (Twin HTLS)* 400 kV D/c line or Jaisalmer (RVPN) S/s - Fatehgarh-III PS (Twin HTLS)* 400 kV D/c line.

** with minimum capacity of 2100 MVA on each circuit at nominal voltage*

 # Tentative Start date of Connectivity at Fatehgarh-IV PS (section-2) is Sep'25 (The scheme is under approval for NCT). Final Start date shall be given after award of ISTS work.

3. Conventional Connectivity Application

It was informed that following 1 No. of Conventional Connectivity application from M/s Renew Solar (Shakti Thirteen) Private Limited was received in the month of Feb'23 with following details:

TABLE 3

Sl. No.	Application ID	Name of the Applicant	Application Type	Submission Date	Region	Project Location	Start Date of Connectivity	Quantum (MW)
1	0010800002	Renew Solar (Shakti Thirteen) Private Limited	Bulk Consumer	21/02/2023	NR	Panipat, Haryana	01/06/2025	100

- It was deliberated that M/s Renew Solar (Shakti Thirteen) Private Limited has applied for Connectivity to the ISTS for 100 MW as a bulk consumer located at Panipat, Haryana with start date of Connectivity as 01/06/2025. As per the application, the bulk consumer entity is located close to the existing 400/220 kV Jind (PG) substation.
- It was enquired from M/s Renew that whether this load is a new load or already connected load to STU/Discom. M/s Renew submitted that this is all new load from Green hydrogen plant in Panipat Industrial area. They also added that their location is more near to Sonepat S/s of Power Grid, however, as there are several ROW constraints near Sonipat (PG) S/s, therefore, they have requested for Connectivity at Jind (PG) substation.
- In view of above, following transmission system was agreed for Connectivity to M/s Renew for 100 MW as Bulk Consumer in Panipat, Haryana:
  - Renew Solar (Shakti Thirteen) Private Limited Load– Jind (PG) 220 kV D/c line – (approx. 64 km) along with associated line bays at both end- Cost about Rs 94 Cr
- M/s Renew enquired that whether they can opt for S/c line instead of D/c line. Regarding this, CTU mentioned that as per the CEA Planning Criteria, N-1 condition is mandatory for conventional Projects. Grid India also confirmed as requirement of D/c line as per the Planning Criteria.
- Further, it was deliberated that there is no provision in the Electricity Act or in the 2009 Connectivity Regulations for construction of a dedicated transmission line by a bulk consumer. The same has also been noted by the Hon'ble Commission and in some specific bulk consumer cases, CERC has directed that the dedicated transmission system to be constructed is to be by the applicant/ any other transmission licensee, with the cost of construction of transmission line for connectivity to ISTS to be borne by applicant.

Further, the Commission said "We are not inclined to issue any general directions in this regard and such matters shall be decided on case-to-case basis depending upon specific circumstances and facts of the case".

- Accordingly, it was agreed to grant Connectivity to M/s Renew as Bulk Consumer at 400/220 kV Jind (PG) Substation for 100 MW of load, however, for implementation modality of the dedicated line in instant case, M/s Renew may also approach CERC for the same.

4. <u>Proposal for grant of LTA to the application received from RE Energy Sources in Feb'23</u>

The following LTA application was received in the month of Feb'23. Details are as below:

TABLE 4

Sl. No.	Application No.	Applicant	LTA Application date (online)	Connectivity Injection Point	Connectivity Application	Drawl Point	Quantum of LTA (MW)	Start Date of LTA sought as per application	End date of LTA
1	0412100025	Prerak Greentech Private Limited	28.02.2023	Bikaner-II PS, Rajasthan	1200003770	Target (NR)	226	01/01/2024	01/01/2049

It was deliberated that Stage-II Connectivity (1200003770, 340 MW) was granted to M/s Prerak Greentech Private Limited at Bikaner-II PS through 220 kV S/c line. 1 No. 220 kV bay at Bikaner-II PS is being implemented under ISTS. Now, M/s Prerak Greentech has applied for LTA of 226 MW on target (NR) basis. Applicant confirmed the same.

It was mentioned that based on the system studies, Rajasthan SEZ Phase-I & II Transmission schemes is already agreed in 2nd and 5th NRSCT meetings held on 13/11/2018 & 13/09/2019 respectively. The required power transfer of 226 MW from above application, from Bikaner-II PS was envisaged under the said system. In addition to this, the transmission system for present LTA, consists of Part-G of Ph-II Tr. System with tentative commissioning schedule of Nov'23, North-West Inter-regional system strengthening scheme (May'24) and 500 MVA 400/220 kV (3rd) ICT at Bikaner-II PS (Tentatively by Nov'24) for LTA.
Details of transmission system for present LTA is mentioned at **Annexure-I.**

Accordingly, it was agreed to grant LTA to M/s Prerak Greentech Solar Private Limited for 226 MW from Bikaner-II PS to NR on target basis from 01/12/2024 (Interim) to 01/01/2049. Final Start date of LTA shall be given upon award of ISTS elements.

5. Proposal for grant of LTA to the applications to be processed from the month of Dec’2022

TABLE 5

Sl. No.	Application No.	Applicant	LTA Application date (online)	Connectivity Injection Point	Connectivity Application	Drawl Point	Quantum of LTA (MW)	Start Date of LTA as per application	End date of LTA
1	SW9999482858-M029_D001_A007-1671556542236	Amplus Ages Private Limited	20.12.2022	Bikaner-II PS, Rajasthan	1200003624	Target (WR)-08 MW Target (ER)-14 MW Target (NR)-20 MW Target (SR)-20 MW	62 (Target)	31/10/2023	30/10/2048
2	SW9990897533-M029_D001_A007-1672319027575	Tepsol Sun Sparkle Private Limited	29.12.2022	Bhadla-III PS, Rajasthan	0212100030	Target (WR)-100 MW Target (NR)-100 MW Target (SR)-100 MW	300 (Target)	01/10/2025	30/09/2050
3	0412100009	Juniper Green Stellar Private Limited	30.12.2022	Fatehgarh-IV PS, Rajasthan	1200003827-100MW, 1200003910-100MW,	Target (WR)-50 MW Target (NR)- 50 MW Target (SR)-50 MW	150 (Target)	30/06/2025	30/06/2050
4	0412100010	Juniper Green Stellar Private Limited	30.12.2022	Fatehgarh-IV PS, Rajasthan	1200003910-100MW, 1200003958-60MW, 0312100010-45MW	Target (WR)-50 MW Target (NR)- 50 MW Target (SR)-50 MW	150 (Target)	30/06/2026	30/06/2051
5	0412100012	Juniper Green Beta Private Limited	30.12.2022	Bhadla-III PS, Rajasthan	0212100027	Target (WR)-50 MW Target (NR)- 50 MW Target (SR)-50 MW	150 (Target)	30/06/2025	30/06/2050
6	0412100022 (Change in Region)	Renew Surya Aayan Private Limited	31.01.2023	Fatehgarh-III PS, Rajasthan	1200002692	KSEB (SR)	300 (PPA/PSA Without Noc)	30/09/2023	29/09/2048

A. <u>Regarding LTA Applications with drawl in Western/Southern Region</u>:

It was deliberated that above LTA applications were discussed in 15th and 16th SR-CMETS meeting held on 30/01/2023 and 28/02/2023 respectively, wherein it was informed that present ATC for import of power from NEW grid to SR is 18900 MW. The ATC is expected to be enhanced to about 23000 MW with the commissioning of transmission scheme viz. additional inter-Regional AC link for import into Southern Region i.e. Warora-Warangal and Chilakaluripeta - Hyderabad - Kurnool 765 kV link. The margins of 20460 MW in ISTS for import of power to SR grid from NEW Grid has already been allocated under different LTA / MTOA granted / agreed for grant with existing / under implementation transmission system / inter-regional links. As the total power transfer requirements are exceeding the expected enhancement in ATC with the additional under implementation inter-regional links, new inter – Regional links shall be required for import of power from NEW Grid to SR and detailed transmission system studies are being carried along with the Western/Southern Region stake-holders for identification same.

Subsequently, LTA granted for 763 MW to M/s Karnataka Power Corporation Ltd. for import of power from NEW Grid (2x800 MW, Godhna TPS in Chhattisgarh) to SR (Target) has been cancelled / revoked vide letter dated 28/02/2023. With consideration of this additional margin (763 MW), study analysis has been carried out and it is observed that all LTA applications received in month of Dec.'2022 **(i.e. Applications at Sl. No. 1 to 5 at Table-5)** having injections in other regions and drawl in SR, for total drawl of 3450 MW may be granted LTA with existing / under implementation transmission system / inter-regional links.

However, for import of additional 2250 MW from NEW Grid to SR from LTA applications received in month of Jan.'2023, additional inter-regional link shall be required. The detailed studies for evolution of additional inter-regional link between NEW grid & SR grid was discussed on 21-22 Feb'23 with SR take holders in SRPC and further, detailed studies are being carried out along with Western/Southern Region stake-holders through joint meeting for finalization of the same.

In view, of above it was agreed to reserve the margins in ISTS for import of power from NEW Grid to SR for total quantum of 3450 MW with the commissioning of transmission scheme with existing / under implementation transmission system / inter-regional links i.e. Warora-Warangal and Chilakaluripeta - Hyderabad - Kurnool 765 kV link. The same was deliberated in the 17th SR-CMETS meeting held on 31/03/2023 also for approval.

Further, during the present meeting, Grid India Ltd. informed that there may be constraints for such drawl of power on SR w.r.t WR-NR inter-regional flows and ATC/TTC may be breached in such scenario. On this, CTU submitted that for such scenario, WR-NR inter-regional strengthening, "North-West Inter-regional system strengthening scheme" is already planned in two phases with implementation schedule of May'24 and the same shall improve WR-NR ATC/TTC & further at the Fatehgarh-IV (section-2) pooling station, LTAs are being considered with Jalore-Mandsaur-Beawar 765 kV D/c Inter-regional corridor. Further, these LTAs shall be effective in the time frame of 2025-26, when all these IR links would be available. However, it was decided that the NR-WR, ATC/TTC shall be reviewed in view the LTA being granted with drawl in WR/SR and all the LTA applications at Table-5, shall be discussed again in NR-CMETS meeting.

6. Proposal for grant of LTA to applications received in other regions for Injection/drawl in Northern region

a) LTA applications received in Western region for drawl in NR

TABLE 6

Sl. No.	Application ID	Name of the Applicant	Submission Date	Quantum of LTA	Start Date of LTA	End Date of LTA	Injection Point	Drawl Point
1.	0431400004	Serentica Renewables India 4 Private Limited	19.01.2023	100 (Firm)	31/12/2024	30/12/2049	Kallam PS, Maharashtra (WR)	Hindustan Zinc Limited, Kankroli S/s, Rajasthan (NR)
2.	0412100015	Serentica Renewables India 4 Private Limited	19.01.2023	100 (Firm)	31/03/2025	30/03/2050	Kallam PS, Maharashtra (WR)	Hindustan Zinc Limited, Kankroli S/s, Rajasthan (NR)
3.	0431400006	Renew Solar Power Private Limited	13-02-2023	75 (PPA/PSA without NoC)	28-06-2023	27-06-2048	RSPPL, Kallam PS, Maharashtra/WR	WBSEDCL, West Bengal/ER

For Applications at Sl. No. 1 & 2: It was noted that the above LTA involves power transfer from WR to NR and no constraints are envisaged in above transfer of power. The detailed transmission system for injection of power in WR was also discussed and agreed in the 16th CMETS-WR meeting held on 27/02/2023. However, regarding applications with drawl at Kankroli Substation (NR), it was deliberated in 16th NR-CMETS meeting that detailed studies need to be carried out. Accordingly, studies were carried out considering shifting of load of M/s Hindustan Zinc limited (HZL) from 220 KV Kankroli (RVPN) S/s to 220 kV level of 400/220kV Kankroli (PG) S/s. From the planning studies, it emerged that there is margin available at Kankroli (PG) S/s for drawl of above 200 MW (100 MW from each application) through 3x315 MVA 400/220 kV ICT of Kankroli S/s. In the meeting, Grid India highlighted that Kankroli ICT level may be rechecked as in peak load scenario, ICT loading are near N-1 limits. However, they shall re-confirm on the same with real time data and share study file with CTU. Subsequently, NRLDC vide their email dated 06/04/2023 informed that Kankroli (PG) has three 315 MVA ICTs and shared their study files. Based on the maximum loading of Kankroli ICTs (215 MW per ICT) and the n-1 permissible loading limit (238 MW per ICT), there is only a 69 MW margin left to draw power through the Kankroli ICTs while complying with the n-1 criteria. Therefore, the above information may be taken into account while granting LTA.

Based on the NRLDC inputs, CTUIL carried out revised studies considering the peak load of Kankroli complex. From the revised studies, loading of Kankroli ICTs are observed to be N-1 non-compliant with drawl of above load (200 MW). Considering above, there is a need of ICT augmentation at Kankroli S/s with 500 MVA ICT (4^{th}) for subject LTA. POWERGRID vide their email dated 14/10/2022 also confirmed the space for required ICT. However, it is also mentioned that in order to accommodate 4^{th} ICT, 50MVAr Bus Reactor needs to be shifted at future bay no. 416 and 4th ICT will be placed at existing location of Bus Reactor-1. 220kV side of ICT will be extended through 220kV Cable/GIB. 220kV side ICT bay will be AIS Type with ICT side termination through 220kV GIB/cable.

Details of the transmission system for present LTA for drawl at Kankroli S/s:

- 1x500 MVA.400/220kV ICT (4th) at Kankroli S/s along with associated 220kV bay
- Shifting of existing 50 MVAr Bus reactor [so as to accommodate ICT] and development of bay for reactor
- Extension of 220kV side of ICT through 220kV Cable/GIB

Implementation Timeframe: 18 months

<u>For Applications at Sl. No. 3:</u> It was deliberated that M/s RSPPL was granted 300MW Stage-II connectivity at Kallam PS based on SECI LOAs dated 04.06.2020 for supply of Round the Clock (RTC) power from 400MW RE Power Projects to various beneficiaries.

Contract Capacity	Project-components	Technology	Individual Capacity at each location (MW)	Location of connectivity
400MW	Sub-Component 1	Solar (400MW) + Battery Energy Storage System (8MWh)	400	ISTS 400/220 kV Fatehgarh III Sub-station, Rajasthan
	Sub-Component 2	Wind	300	ISTS 400/220 kV Kallam (Osmanabad) substation, Maharashtra
	Sub-Component 3	Wind	300	ISTS 400/220 kV Koppal substation, Karnataka
	Sub-Component 4	Wind	300	ISTS 400/220 kV Gadag substation, Karnataka

The project configuration was informed as under:

M/s RSPPL has already been granted LTA for 300MW to Target beneficiaries in ER: 75MW, NR: 150MW & WR: 75MW with following transmission system:

- Establishment of 400/220 kV, 2X500 MVA Kallam PS
- LILO of both circuits of Parli (PG) – Pune (GIS) 400kV D/c line at Kallam PS

The above application is for the change in region for 75MW (45MW-NR 30MW-WR to 75MW-WBSEDCL, ER) (out of 300MW) LTA already granted to RSPPL against app. no. 1200003270. Applicant has submitted request for relinquishment of earlier granted 75MW LTA w.e.f 28.06.2023. Further, applicant has also requested to firm-up its balance LTA granted earlier on target region basis as follows:

Sl. No.	Region	Target quantum as per previous grant	Quantum after change in region	Firmed-up beneficiary	Firmed-up Quantum requested
1	Northern Region	150 MW	105MW	Uttarakhand Power Corporation Ltd – 75MW Northern Railway (Uttar Pradesh) – 30MW	105 MW
2	Eastern Region	75 MW	150MW	West Bengal State Electricity Distribution Company Ltd – 75MW India Power Corporation Ltd – 75MW	150 MW
3	Western Region	75 MW	45MW	Northern Railway (Madhya Pradesh) – 45MW (PPA signed by Northern Railway for Western Central Railway)	45 MW

PPA for 400MW between SECI "Renew Surya Roshni Pvt. Ltd." which is the 100% subsidiary of "RENEW SOLAR POWER PRIVATE LIMITED" has been submitted whereas PSA between SECI and following drawee entities with quantum has been submitted:

- 100MW with INDIAN RAILWAY
- 100MW with India Power Corporation Ltd
- 100MW with Uttarakhand Power Corporation Ltd
- 100MW with West Bengal State Electricity Distribution Company Ltd

NoC has been submitted from MPPTCL (60MW drawl) and UP STU (40MW drawl) against Indian Railway quantum. However, other NoCs have not been submitted.

From the above, it may be noted that the quantum mentioned in PSA is not matching with quantum mentioned in request for firming up of beneficiaries.

Further, CTU enquired about the drawl/injection pattern of generation w.r.t Connectivity & LTA applications in NR, SR & WR. Applicant clarified that their application in various regions is based on SECI LOAs dated 04.06.2020 for supply of Round the Clock (RTC) power from 400MW RE Power Projects to various beneficiaries. POSOCO submitted that as this first of its kind of project, therefore, its scheduling shall be challenging in practical situation for Grid Controller.

In view of this, it was deliberated that the above LTA grant was also discussed in the 17th WR-CMETS meeting held on 30/03/2023, wherein, it was decided that the above matter shall be discussed in a separate meeting among CTU, Grid India, M/s Renew & RLDCs. Based on the outcome of this meeting, the application shall be taken up for discussion in all respective regions for grant of LTA.

b) LTA applications received in Southern region for drawl in NR:

Details of LTA applications (with injection in SR and drawl in other regions) received in Feb' 2023, in conformity with CERC Regulations, are given below:

Sl.	Application ID	Name of the Applicant	Submission Date	Quantum of LTA	Start Date of LTA	End Date of LTA	Injection Point	Drawl Point
1.	0451100014	Renew Solar Power Pvt Ltd.	13.02.2023	75 MW (PPA / PSA without NoC)	30.06.2023	29.06.2048	Gadag PS, Karnataka	• WBSEDCL(ER): 75 MW
2.	0451100015	Renew Solar Power Pvt Ltd.	13.02.2023	75 MW (PPA / PSA without NoC)	31.03.2023	30.03.2048	Koppal PS, Karnataka	• WBSEDCL(ER): 75 MW
3.	0451100016	Renew Surya Ojas pvt Ltd.	13.02.2023	100 MW (PPA / PSA without NoC)	31.03.2023	30.03.2048	Koppal PS, Karnataka	• HPPC (NR): 50 MW • Goa State (WR): 50 MW

I. M/s Renew Solar Power Pvt. Ltd. (RSPPL) has already been granted LTA for 300 MW (application no. 1200003268) with injection at Gadag PS and drawl in NR (150 MW), ER (75 MW) and WR (75 MW) on target region basis. Presently, vide letter dated 26.12.2022 (as part of LTA applicant no. 0451100014), applicant has sought for change in quantum of LTA in regions as well as firm up of beneficiaries. Details of same are given below.

Beneficiaries as per earlier grant	Firmed up beneficiaries as per present request
NR (Target) : 150 MW	• Uttarakhand Power Corporation Ltd. : 75 MW • Northern Railway (Uttar Pradesh) : 30 MW
WR (Target) : 75 MW	• Northern Railway (Madhya Pradesh) : 45 MW
ER (Target) : 75 MW	• West Bengal State Electricity Distribution Company Ltd. (WSEDCL) : 75 MW • India Power Corporation Ltd. : 75 MW

Further, applicant has submitted that change in LTA from NR to WSEDCL (ER) for 45 MW and WR to WSEDCL (ER) for 30 MW is effective from 30.06.2023 and there is no change in LTA operationalization date. The application was also discussed in CMETS-SR meeting held on 31.03.23.

II. M/s Renew Solar Power Pvt. Ltd. (RSPPL) has already been granted LTA for 300 MW (application no. 1200003267) with injection at Koppal PS and drawl in NR (150 MW), ER (75 MW) and WR (75 MW) on target region basis. Presently, vide letter dated 26.12.2022 (as part of LTA applicant no. 0451100015), applicant has sought for change in region as well as firm up of beneficiaries. Details of same are given below.

Beneficiaries as per earlier grant	Firmed up beneficiaries as per present request
NR (Target) : 150 MW	• Uttarakhand Power Corporation Ltd. : 75 MW • Northern Railway (Uttar Pradesh) : 30 MW
WR (Target) : 75 MW	• Northern Railway (Madhya Pradesh) : 45 MW
ER (Target) : 75 MW	• West Bengal State Electricity Distribution Company Ltd. (WSEDCL) : 75 MW • India Power Corporation Ltd. : 75 MW

Further, applicant has submitted that change in LTA from NR to WSEDCL (ER) for 45 MW and WR to WSEDCL (ER) for 30 MW is effective from 31.03.2023 and there is no change in LTA operationalization date. The application was also discussed in CMETS-SR meeting held on 31.03.23.

It was also informed that as per submitted documents PPA has been signed between M/s SECI and M/s Renew Surya Roshni Pvt. Ltd., which is the 100% subsidiary of M/s Renew Solar Power Pvt. Ltd., for supply of 400MW. Further, PSA has been signed between M/s SECI and following drawee entities.

- Indian Railway (100MW : UP for 40 MW & MP for 60MW))
- India Power Corporation Ltd. (100MW)
- Uttarakhand Power Corporation Ltd. (100MW)
- West Bengal State Electricity Distribution Company Ltd. (WBSEDCL) (100MW)

NoC has been submitted from MPPTCL (60MW drawl) & UP STU (40MW drawl) against drawl by Indian Railway and from West Bengal State Electricity Transmission Company Ltd. (WBSETCL) for 100 MW drawl (WBSEDCL). However, NoCs for other drawees beneficiaries have not been submitted. Further, it may be noted that the quantum mentioned in PSA and NoC is not matching with quantum mentioned in request for firming up of beneficiaries.

M/s Renew Power Pvt. Ltd. vide email dated 27.03.2023, clarified the followings with respect to projects.

A) Clarification regarding project configuration & PPA/PSA mismatch :

- Bidding Company – ReNew Solar Power Private Limited (RSPPL)
- SPV for the project – ReNew Surya Roshni Private Limited (RSRPL)
- Project Configuration – 400 MW Solar+8MWh BESS at Fatehgarh III in Rajasthan and 900 MW Wind (300 MW at Koppal & 300 MW at Gadag in Karnataka and 300 MW at Kallam in Maharashtra)
- PPA Quantum : 400 MW
- PSA Quantum: 100 MW UPCL (UK); 100 MW IPCL (WB), 100 MW WBSEDCL (WB), 100 MW Indian Railways (40 MW Northern Railway -UP drawl and 60 MW Western Central Railways – MP drawl)
- Proposed LTA Mapping : For 400 MW injection at Fatehgarh III, LTA mapping will be as per PSA quantum, however, for 300 MW individual injection at Koppal, Gadag & Kallam, LTA mapping will be 75% of PSA quantum as given under:

Applicant Name	Injection Point	Connectivity & LTA Quantum (MW)	LTA Mapping
ReNew Surya Roshni Pvt. Ltd.	Fatehgarh-III	400	• NR : 140 MW (UPCL – 100 MW; IR with drawl at UP – 40 MW) • ER : 200 MW (WBSEDCL – 100 MW; IPCL – 100 MW) • WR : 60 MW (IR with drawl at MP – 60 MW)
ReNew Solar Power Pvt Ltd.	Koppal	300	• NR : 105 MW (UPCL – 75 MW; IR with drawl at UP – 30 MW) • ER : 150 MW (WBSEDCL – 75 MW; IPCL – 75 MW) • WR : 45 MW (IR with drawl at MP – 45 MW)
ReNew Solar Power Pvt Ltd.	Gadag	300	
ReNew Solar Power Pvt Ltd.	Kallam	300	

B) Clarification regarding scheduling of power:

- As per PPA, RSPPL need to maintain min. 70% monthly and 80% annual CUF for 400 MW contracted capacity and accordingly, project will have 1300 MW (Solar – 400 MW; Wind – 900 MW) of installed capacity with injection at 4 ISTS points in 3 states.
- Scheduling of power to the Buying Entities under the PPA will not go beyond 400 MW in any given time block cumulatively from all the Injection ISTS points (Fatehgarh III, Koppal, Gadag, Kallam), however, quantum of injection at Fatehgarh III can go upto 400 MW and for Koppal, Gadag & Kallam upto 300 MW at each ISTS point to schedule 400 MW power & to meet CUF requirements under PPA.

Matter was deliberated in detail wherein SRLDC submitted that as this project (RTC) is first of its kind of project, therefore, its scheduling shall be challenging in practical situation for Grid Controller.

Above matter was also deliberated in the 17th WR-CMETS meeting held on 30/03/2023, wherein, it was decided that the above matter shall be discussed in a separate meeting among CTU, Grid India, M/s Renew & RLDCs. Based on the outcome of this meeting, the application shall be taken up for discussion in all respective regions for grant of LTA.

After detail deliberation it was decided to take-up the matter in separate meeting as mentioned above.

III. M/s Renew Surya Ojas Pvt. Ltd. (RSOPL) has already been granted LTA for 300 MW (application no. 1200002993) with injection at Koppal PS and drawl in NR (100 MW), ER (100 MW) and WR (100 MW) on target region basis. Presently, vide letter dated 26.12.2022 (as part of LTA applicant no. 0451100016), applicant has sought for change in region as well as firm up of beneficiaries. Details of same are given below.

Beneficiaries as per earlier grant	**Firmed up beneficiaries as per present request**
NR (Target) : 100 MW	Haryana Power Purchase Centre (HPPC) : 150 MW
WR (Target) : 100 MW	Goa : 150 MW
ER (Target) : 100 MW	-

Further, applicant has submitted that change in LTA from ER to HPPC (NR) for 50 MW and ER to Goa (WR) for 50 MW is effective from 31.03.2023 and there is no change in LTA operationalization date. Applicant has also submitted copy of PPA and PSA without NoC. The matter was deliberated in 17th CMETS-WR meeting and the above was agreed subject to agreement in the CMETS-SR meeting and as per the GNA Regulations.

After detailed deliberations, it was agreed that for the proposed power transfer of 150 MW to HPPC in NR, there is adequate margin in the inter-regional corridors and no constraints are envisaged for the same for above LTA.

c) <u>LTA applications received in Eastern region for drawl in NR:</u>

The following LTA application was received from East Central Railway (ECR) in Feb 2023 under CERC Connectivity Regulations, 2009 with injection in ER and drawl in NR and ER:

Sl. No.	Application ID	Name of Applicant	Submission Date	Quantum of LTA	Start Date of LTA	End Date of LTA	Generation/ Injection Point	Drawl Point
1.	0440100003	East Central Railways	28-02-2023	25 (Firm)	01-04-2023	31-01-2035	BRBCL, Bihar (ER)	East Central Railway, Bihar (ER) – 20MW Delhi (NR) – 5MW

The applicant has once again requested for redistribution of already operational 819 MW LTOA w.e.f. 01-04-2023 upto 31-01-2035 as detailed below.

Region	State	Existing LTOA quantum (MW) w.e.f. 01-01-2022	Change sought (MW) in Dec 2022 appl.	Revised LTOA quantum proposed in appl. No. 1672316908118 dtd. 29-12-2022	Change sought (MW) in Feb 2023 appl.	Revised LTOA quantum proposed in appl. No. 0440100003 dtd. 28-02-2023
ER	Bihar – STU	100	0	100	0	100
	Bihar – ISTS	0	0	0	+20	**20**
	DVC	110	0	110	0	110
Sub-Total (ER)		**210**	**0**	**210**	**+20**	**230**
WR	Maharashtra	120	0	120	-25	**95**
	Madhya Pradesh	209	0	209	0	209
Sub-Total (WR)		**329**	**0**	**329**	**-25**	**304**
NR	Uttar Pradesh: ISTS interconnection points with Railway network	***55***	***-30***	***25***	0	25
	Uttar Pradesh: ISTS interconnection points with UP-STU network	***95***	***-20***	***75***	0	75
	Haryana	55	0	55	0	55
	Punjab	35	0	35	0	35
	Rajasthan	10	0	10	0	10
	Delhi	15	0	15	+5	**20**
Sub-Total (NR)		**265**	**-50**	**215**	**+5**	**220**
SR	Karnataka	***10***	***+50***	***60***	0	60
Sub-Total (SR)		**10**	**+50**	**60**	**0**	**60**
NER	Assam	5	0	**5**	**0**	**5**
Sub-Total (NER)		**5**	**0**	**5**	**0**	**5**
Total		**819**	**0**	**819**	**0**	**819**

The application was discussed & agreed for grant during 17th ER-CMETS meeting held on 29/03/23.

Based on the studies, it was deliberated that for the proposed power transfer of 20MW to ECR at Sasaram and 5MW to NR (Delhi), there is adequate margin in the inter-regional corridors and power flow on the lines is within acceptable limits. Thus, it was agreed to grant LTA on existing transmission system.

B. ISTS Expansion

i. Augmentation of 1x500 MVA (4th) ICT at 400/220 kV Nallagarh substation

In the Quarterly operational feedback of October 2022, Grid-India highlighted the issue of 'N-1' non-compliance of loading of ICTs at Nallagarh (PG) S/s. Grid-India informed that currently with 3 no. of 315 MVA ICTs, loading of ICTs was more than 'N-1' compliance limit (>750 MW) during high drawl by HP and Punjab. Grid-India recommended that new ICT may be planned to relieve loading. Subsequently, the same was also discussed in the 62nd NRPC held on 31.01.2023.

The loading pattern of Nallagarh ICTs of past one year is as below:

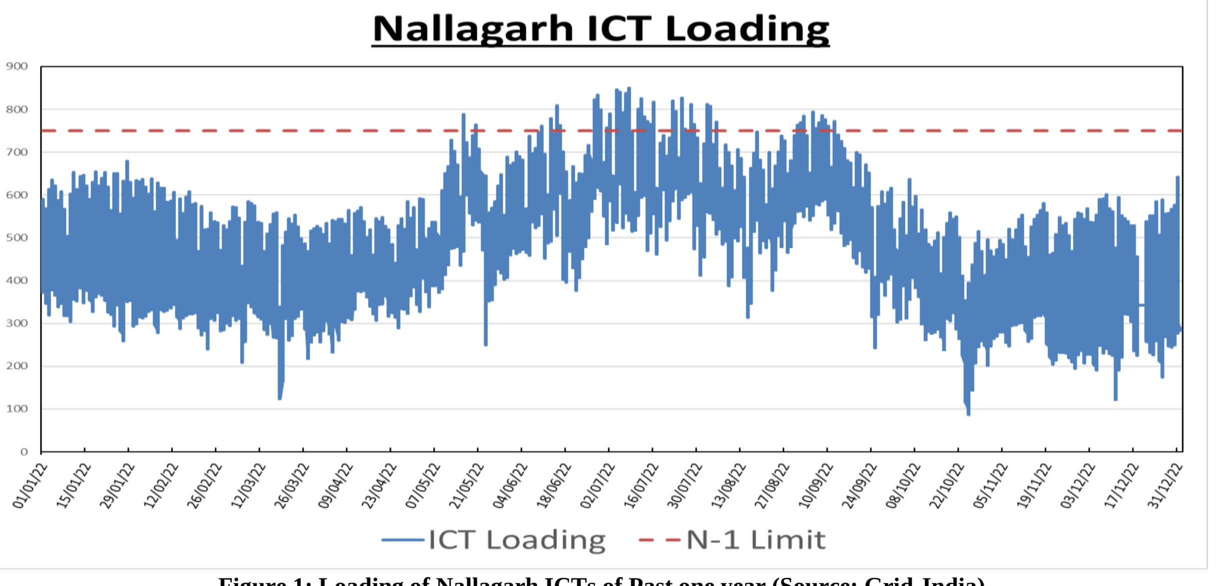

Figure 1: Loading of Nallagarh ICTs of Past one year (Source: Grid-India)

From the loading pattern of Nallagarh ICTs (3x315 MVA), it is observed that the loading is above N-1 limit for sufficient duration of time from May'22 to September-22 period. Loading of ICTs has reached maximum of about 850MW sometimes in July'22 and Aug'22.

POWERGRID vide mail dated 10.03.2023 confirmed that the space is available at Nallagarh S/s for installation of 4th ICT. However, 220 kV Bay along with its interconnection with 4th ICT shall be through GIS.

HPPTCL stated that prim facie there is requirement to augment Nallagarh ICT, however they will confirm it after their management approval. PSTCL also stated that they will revert back on requirement to augment Nallagarh ICT.

NRLDC stated that ICT augmentation at Nallagarh is required on urgent basis. Grid-India vide mail dated 06/04/23 informed that 400/220kV ICTs at Nallagarh S/s is N-1 non-compliant during high load scenario in Punjab restricting the Punjab's import Capability (ATC) from Grid. CEA also agreed for the above requirement.

Subsequently HPPTCL vide mail dated 13.04.2023 and PSTCL vide mail dated 19.04.2023 provided their consent for augmentation of 400/220 kV ICTs at Nalagarh.

Considering above consents as well as future load growth in Punjab, Chandigarh & HP as well as to satisfy the N-1 criteria of loading of ICTs, it was agreed to implement one no. of 400/220kV, 500 MVA ICT (4th) at Nallagarh S/s under ISTS with following scope:

- 400/220 kV, 500 MVA ICT(4th) at Nallagarh (PG) S/s
- Implementation of associated ICT bays (1 no. of 400 kV and 1 no. of 220 kV (GIS bay)) at Nallagarh (PG) S/s
- GIS duct for interconnection of 220 kV bay with ICT

Implementation Timeframe: 18 months

ii. <u>Intra State transmission schemes of PTCUL involving inter connection with ISTS elements</u>

CEA vide mail 26.12.22 informed that PTCUL has submitted the proposal for following Intra State transmission schemes which involve inter-connection with ISTS elements.

Part A:

- Establishment of 400/220 kV (2X315 MVA) AIS Substation Landhora
- LILO of one circuit of 400 kV D/c Kashipur (PTCUL) – Roorkee/Puhana(PG) line (LILO length: 3 km) at Landhora
- LILO of 220 kV Roorkee - Manglore (proposed) line (LILO length: 25 km) at Landhora.

Part B:

- Establishment of 132 kV/33 kV (2x20 MVA) GIS Substation Lohaghat
- Pithoragarh (PGCIL) – Lohaghat 132 kV D/c line (Line length: 39.33 km)

In this regard, CTU vide mail dated 08.02.23 informed that the above proposal was examined in joint studies of CTU and PTCUL on 12th & 13th Jan'23 as part of PSSE file updation for Rolling plan 2027-28.

PTCUL informed that presently 220 kV Roorkee (PTCUL) substation is getting power mainly from 220 kV S/C Roorkee-Puhana line. Further 220 kV Roorkee is feeding 03 No. of existing 132 kV substations Manglore, Laksar, Bhagwanpur in radial mode. There is constraint of power flow from 220 kV Roorkee-Nara (UP) line. Roorkee and nearby area are having high industrial load demand. Transmission system strengthening of Roorkee area is required for increasing the reliability of power supply, meeting N-1 contingency criteria at 220 kV level and meeting the future load growth.

PTCUL also informed that 2 no. of 132 kV bays at Pithoragarh(PG) which are already commissioned will be utilized for termination of Lohaghat D/c line.

In view of high voltage scenario in NR, CTU recommended that one no. of 125MVAr (420kV) bus reactor at 400/220kV Landhora S/s along with space provision of additional bus reactor (2nd) may be considered as part of above transmission scheme by PTCUL.

Further, CTU also informed that implementation of the above scheme shall be under the scope of PTCUL. PTCUL may provide the timeline for implementation of above transmission scheme.

PTCUL informed that, there is space available at Landhora S/s and 125 MVAr reactor will be planned accordingly. PTCUL also informed that Lohaghat S/s is already under tendering and Landhora S/s is yet to be tendered. Lohaghat S/s along with associated system are expected to be commissioned around Dec'25 while Landora S/s and associated system is expected by March'26.

NRLDC vide mail dated 6th April' 23 informed that for the proposed plan to commission the 400/220kV Landhora Substation, with the LILO of 400kV Kashipur-Roorkee line at the 400kV Landhora and the 220kV Roorkee-Manglore line at the 220kV Landhora, base case shared by CTU is found to be in order. This scheme is expected to mitigate the present overloading of the 220kV Rookee-Roorkee line. After deliberations, the above proposal of PTCUL was agreed.

iii. <u>Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-3 :2 GW) (Ramgarh Complex)</u>

It was deliberated that Rajasthan REZ Ph-III (20GW) Transmission scheme was agreed in 5th NCT meeting. EHVAC transmission system for 14GW RE potential is already under advance stage of bidding whereas for evacuation of balance 6GW RE potential, HVDC transmission scheme was agreed in 9th NCT meeting and under bidding. As part of above Phase-III scheme, Bhadla-3 PS is being established for integration of its 6.5GW RE potential. Further, Ramgarh PS shall also be connected with Bhadla-3 for evacuation of RE power from Ramgarh Complex (Ph-III potential: 2.9GW). In order to facilitate evacuation of 9.4GW RE power from Ramgrarh/Bhadla-III PS (6.5GW+2.9GW) from Bhadla-3 onwards, 765kV Bhadla-3 - Sikar-II D/c line with implementation schedule of Dec'24 (Tentative) [for 2.9GW power transfer requirement] as well as 6GW HVDC corridor ($\pm$800kV Bhadla (HVDC) - Fatehpur(HVDC)) with implementation schedule of Dec'26 (42 months schedule) is being implemented as part of Ph-III scheme.

It was stated that at present, St-II Connectivity for about 2.75 GW RE and LTA of 2.6GW, against the potential of 2.9 GW (under Ph-III), is already received/granted at Ramgarh PS. Recently M/s Adani vide letter dated 27.02.23 submitted their dissent for GNA transition for their 1.5GW solar project in Ramgarh PS as per CERC (Connectivity & GNA to the ISTS) regulation 2022. However, above dissent by Adani will be considered upon completion of GNA transition process. Therefore, after accounting for above dissent, power transfer requirement (LTA) from Ramgarh will be 1.1GW (2.6GW-1.5GW). St-II Connectivity for about 2.95 GW and LTA of about 0.95GW, against the potential of 6.5 GW (under Ph-III), is already received/granted at Bhadla-3 PS.

Beyond Bhadla-3 PS, Ph-III EHVAC system i.e. 765kV Bhadla-3-Sikar-II D/c will be utilized for power transfer of 2.9GW RE potential of Ramgarh/Bhadla-3 PS. Based on above, LTA application for 2.05GW (1.1GW at Ramgarh + 0.95GW at Bhadla-3) is already processed/under process and balance 0.85GW (2.9GW-2.05GW) will be considered based on receipt of applications on above EHVAC Corridor.

Further, Renewable Energy Zones (REZs) were identified by MNRE/SECI with a total capacity of 181.5 GW for likely benefits by the year 2030 in eight states, which includes 75 GW REZ potential in Rajasthan comprising of 15 GW Wind and 60 GW Solar. In this regard a Committee on Transmission Planning for RE was constituted by MOP for planning of the requisite Inter State Transmission System required for the targeted RE capacity by 2030. As part of committee report "Transmission system for integration of over 500GW capacity by 2030" as well as MNRE/SECI inputs, a Comprehensive transmission plan for evacuation of

75GW RE potential from Rajasthan is evolved comprising 10 GW RE potential (Wind: 4 GW, Solar: 6 GW) along with 3GW BESS(net evacuation 5 GW) at Ramgarh in Jaisalmer complex by 2030. Out of 10 GW potential by 2030, in its Ph-I (by 2025) 3GW potential (Wind: 2GW, Solar: 1GW) was informed at Ramgarh.

Table 2

S.No	Pooling Station	Total RE Potential (by 2030) as per committee report		RE Potential (Ph-I of report: by 2025)	Net (Max) Dispatch Considered by 2025 (GW)
		Source	Capacity (GW)	Capacity (GW)	
1	Ramgarh PS*	Wind	4	2	**2**
		Solar	6	1	
		BESS	3	-	

Solar Dispatch considered @100%, Wind dispatch considered @50% in June Solar Max Scenario

It was deliberated that considering above 2GW additional potential at Ramgarh PS as part of above committee report (beyond Ph-III 2.9GW potential), there is a requirement of an additional 765kV Ramgarh-Bhadla-3 (2^{nd}) D/c line which is proposed in the present agenda. Considering the time schedule for development of above potential (2GW) in Ph-I of report (by 2025) and time schedule of HVDC system beyond Bhadla-3 (Dec'26), there is a need of additional corridor from Bhadla-3 onwards also for which 765kV Bhadla-3 PS - Bikaner-III PS D/c line is proposed. With the development of Ramgarh potential, Ramgarh-Bhadla-3-Bikaner-III 765KV corridor will be utilized. In case potential of Bhadla-3 gets developed before Ramgarh potential, in the shorter timeframe i.e. 2025-26, 765kV Bhadla-3 PS - Bikaner-III PS D/c corridor will only be needed. Further this link would also enhance the short circuit strength Bhadla-3 complex.

As part of above committee report, additional 3 GW RE injection at Ramgarh PS and 2GW RE injection at Bhadla-III/Bhadla-IV PS is envisaged beyond 2025 (Ph-II/III), which will utilize the planned HVDC corridor beyond Bhadla-3 as it will match implementation schedule of generation and transmission i.e. in 2026-27 timeframe.

Considering evacuation requirement from Ramgarh/Bhadla-3 PS, system studies along with study results discussed in the meeting. It was deliberated that in Feb solar maximized scenario (revised case) loading of 400kV RAPS-Shujalpur D/c line is marginally higher (about 950MW) in N-1 contingency. The loading of above line will be reviewed with progress of RE generation projects at Rajasthan and strengthening requirement will be identified later, if required. The line loadings of other lines in above RE complex were found to be in order as well as voltages were within limits on all substations.

The proposed comprehensive Transmission scheme in ISTS is as under:

Proposed Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-3: 2 GW)

1. 765 kV Ramgarh PS - Bhadla-3 D/c line(2nd) along with 240 MVAr switchable line reactor for each circuit at Ramgarh end (~200 km)
2. 765 kV Bhadla-3 – Bikaner-III D/c line along with 240 MVAr switchable line reactor for each circuit at Bhadla-3 end (~150 km)
3. 400 kV Bareilly(765/400kV) – Bareilly(PG) D/c line (Quad) (2nd) (~2 km)
4. Augmentation with 2x1500 MVA, 765/400 kV ICT at Ramgarh PS (In addition to 3x1500MVA ICT in Phase-III Scheme)
5. Augmentation with 3x500 MVA 400/220 kV ICT at Ramgarh PS (In addition to 2x 500MVA ICT in Phase-III Scheme)
6. 5 nos. of 220 kV line bays at Ramgarh PS (for RE connectivity)
7. Augmentation with 1x1500 MVA, 765/400 kV ICT at Bhadla-3 PS (3rd)
8. Augmentation with 1x1500 MVA, 765/400 kV ICT at Bareilly (765/400kV) S/s (3rd)
9. 220kV Sectionalization bay (1 set) along with BC (2 nos.) and TBC (2 nos.) at Ramgarh PS

Note : 400/220kV ICTs and 220 kV line bays (for RE connectivity) at Bhadla-3 PS taken up as per requirement under Ph-III, Part B1/B2 packages (as agreed in 5th NCT Meeting)

Above comprehensive scheme is further segregated into 2 phases based on the development of Ramgarh or Bhadla-3 RE potential

A. With development of RE potential at Ramgarh PS: 2 GW [(1 GW (Solar) + 2GW (Wind)]

1. 765 kV Ramgarh - Bhadla-3 D/c line(2nd) along with 240 MVAr switchable line reactor for each circuit at Ramgarh end (~200 km)
2. 765 kV Bhadla-3– Bikaner-III D/c line along with 240 MVAr switchable line reactor for each circuit at Bhadla-3 end (~150 km)

3. 400 kV Bareilly(765/400kV) – Bareilly(PG) D/c line (Quad) (2nd) (~2 km)
4. Augmentation with 2x1500 MVA, 765/400 kV ICT at Ramgarh PS (In addition to 3x1500MVA ICT in Phase-III Scheme)
5. Augmentation with 3x500 MVA 400/220 kV ICT at Ramgarh PS (In addition to 2x 500MVA ICT in Phase-III Scheme)
6. 5 nos. of 220 kV line bays at Ramgarh PS (for RE connectivity)
7. Augmentation with 1x1500 MVA, 765/400 kV ICT at Bareilly (765/400kV) S/s (3rd)
8. 220kV Sectionalization bay (1 set) along with BC (2 nos.) and TBC (2 nos.) at Ramgarh PS

B. <u>With development of RE potential at Bhadla-3 PS : For 2 GW power transfer requirement</u>

1. 765 kV Bhadla-3 – Bikaner-III D/c line along with 240 MVAr switchable line reactor for each circuit at Bhadla-3 end (~150 km)
2. 400 kV Bareilly(765/400kV) – Bareilly(PG) D/c line(2nd) (Quad) (~2 km)
3. Augmentation with 1x1500 MVA, 765/400 kV ICT at Bhadla-3 PS (3rd)
4. Augmentation with 1x1500 MVA, 765/400 kV ICT at Bareilly (765/400kV) S/s (3rd)

Note : 400/220kV ICTs and 220 kV line bays (for RE connectivity) at Bhadla-3 PS taken up as per requirement under Ph-III, Part B1/B2 packages (as agreed in 5th NCT Meeting)

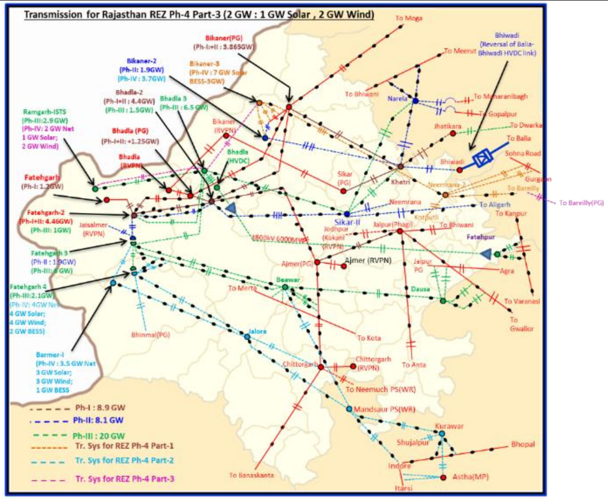

Fig : Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-3) (Ramgarh Complex)

NRLDC vide mail dated 6th April' 23 as well as in meeting asked to share Non Solar base case to review the night time operation. NRLDC also stated that some of the intra state thermal generation (Kawai, Kalisindh, Chhabra, Suratgarh) units may be kept on with minimal dispatch so that same may be available during evening peak hours to meet the Rajasthan demand and to avoid limitation of ATC/TTC of Rajasthan. CTU stated that if thermal units kept on even with lesser dispatches in June solar maximized scenario, intra state generation of Rajasthan (conv+ RE) will be equal/higher than state load and intra state surplus power may have to be dispersed outside Rajasthan, even after ISGS allocation is kept minimum. It was decided that in view of ramping requirement as well as to meet the evening peak load, some of the thermal units may be kept on with minimal dispatch in revised file.

NRLDC also highlighted that few buses (Bhiwadi and other HVDC buses) has STATCOM in study file, CTU stated that they will replace the same with equivalent filter capacity at HVDC stations as per NRLDC data in revised cases. NRLDC also indicated overloading of few lines i.e. 400kV Bhiwani(PG)-Bhiwani(BBMB), Agra (UP)-Agra(PG) etc in study files. CTU stated that Loading on some of 400kV lines is critical in N-1 contingency due to load growth and change in LGB over the time. Same was also highlighted in Rolling plan report-ISTS (2027-28). Proposal of suitable strengthening will be taken up separately with real time loading pattern and load growth in future.

POSOCO stated that studies may also be carried out considering unity power factor or 0.99 pf for solar plants in NR with minimal reactive power support. CTU stated that studies at unity pf may increase the system reactive compensation requirement and it is not defined any applicable standard, however as per Manual on Transmission Planning Criteria 2023, reactive support of ±20% (~0.98 pf) for Wind / Solar / BESS may be taken for the purpose of load flow studies at planning stage. In view of that, intermediate approach in between most optimistic (0.95 pf) and pessimistic scenario (unity) is considered with RE (solar) generator pf of 0.98 in studies.

GRID-India enquired about status to place Synchronous condenser (SYNCON) in Rajasthan. CTU stated that discussion with OEMs is going on at present and once inputs is received from all OEMs planning of synchronous condenser may be taken up in association with grid-India.

GRID-India stated that they have calculated SCR considering IBR contribution (Xs=1) incl. contribution from candidate RE bus (on which SCMVA calculated). From calculation, it emerged that SCR in most of RE pooling stations in western Rajasthan is low (<5). CTU stated that as per present technical standard Short Circuit Ratio at the interconnection point where the generating resource is proposed to be connected shall not be less than 5. However as per international standard/practices shared to GRID-INDIA by CTU, SCR is recommended to 3 which is available in western Rajasthan, therefore deliberation is required for SCR value (3 or 5) for which matter has been referred to CEA also.

GRID-India stated that in real time issues are already faced in terms of transient over voltage during switching as well as power oscillations. CTU stated that real time issues are solely not accounted due to low SCR, as these are also related with inverter control setting and control interaction between various Inverter based resources. Effect of coordinated inverter settings and operation in various modes of RE plants has already shown better results in real time conditions as per the information provided by Grid-india during various meetings.

Grid-India stated that due to low SCR, impact of controller is not up to the required level. With weak grid (low SCR), voltage changes are higher in any event/switching, however if SCR is increased in Grid, with same setting of Inverter controller, issues may not come. CTU stated that as per global practices, RE generators shall have to tune their Inverter setting in line with the available SCR of POI. AEMC (Australian EMC) rules also require connecting IBR generators to demonstrate that they can operate stably down to a minimum short circuit ratio (SCR) of 3.0.

CTU stated that in 12th NCT meeting held on 24.03.23, it was decided that a small group comprising of officers from CEA, CTUIL and GRID-India be constituted to suggest a methodology to compute the system strength (SCR).

CTU also mentioned that based on committee decision on SCR computation methodology, wherever SCR is low, additional system requirement (corridors/devices i.e. synchronous condenser) shall be evolved if required.

CTU stated that with installation of small size SYNCON (Synchronous condenser) may not improve the SCR up to required level and therefore large size SYNCON would require more space at ISTS pooling stations. Therefore, planning is to be done in holistic manner with techno economical solution to improve SCR. In this regard a meeting will be convened shortly by CEA for deliberation on methodology to calculation SCR. Subsequently, in the meeting held on 11.04.23 by CEA, different methodologies for calculation of SCR was discussed. In the meeting, it was also decided that further deliberation is required to decide on SCR calculation methodology in Indian context.

Grid-India stated that at present NR to WR actual flow is exceeding the ATC limit during peak solar hours. Infact the existing LTA+MTOA schedule is more than ATC on NR to WR path during peak solar hours. Therefore, the proposal to grant additional LTA on the NR to WR path should be reconsidered, considering the additional margin that will be created by the commissioning of future transmission elements on this path. Grid-India suggested that CTU share this information (base case with timelines), so that proposal can be reviewed.

CTU indicated that various corridor towards Western region, HVDC reversal to ER, additional corridor from Rajasthan towards NR load centers, HVDC corridor are already under implementation/planned which will enhance NR-WR ATC/TTC. Same is also uploaded on CTUIL website for 2024-25. CTU agreed to share the PSS/E file for ATC/TTC computation. It was decided that transmission scheme proposal will be reviewed and taken up in subsequent CMETS-NR meeting considering observations from GRID-India.

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Annexure-A

The following provisions shall be applied with respect to grant of connectivity to **RE projects** to the ISTS Grid:

i. The grant of Stage-I Connectivity shall not create any vested right in favor of the grantee on ISTS infrastructure including bays. Stage-I Connectivity grantee shall be required to update the quarterly progress of development of their generation project and associated transmission infrastructure/dedicated line as per format **RCON-I-M** by 30th day of June and 31st day of December of each year. Further, Stage-I Connectivity grantees who fail to apply for Stage-II Connectivity within 24 months from grant of Stage-I Connectivity shall cease to be Stage-I grantee and their Application fees shall be forfeited.

ii. If the capacity of the Stage-I Connectivity location is allocated to other Stage-II grantees, the balance Stage-I grantees shall be allocated Stage-II Connectivity to an alternate location.

iii. For optimization of ROW, it is proposed that transmission towers of various dedicated lines, up to 2-3 km periphery from the entry of the pooling station may be of **Multi-circuit type**. The generation developers/applicants associated with the various pooling stations may coordinate amongst themselves for implementation of such span/stretches of the dedicated connectivity lines with M/C towers.

iv. The grant of Stage-I / Stage-II Connectivity shall not entitle an applicant to interchange any power with the grid unless it obtains LTA/MTOA/STOA for power transfer requirements. As grant of LTA may require transmission system strengthening, the applicants are advised to apply for LTA immediately, to enable timely transfer of power to the beneficiaries. Also as per the prevailing Transmission Planning Criteria of CEA, “N-1” contingency criteria may not be applied to the immediate connectivity of wind/solar farms with the ISTS/Intra State grid.

v. Stage-II Connectivity Grantee shall sign the Transmission Agreement for Connectivity and submit the Connectivity Bank Guarantee (**Conn-BG1 and Conn-BG2**) to CTU within 30 days of issue of intimation. No extension of time shall be granted and in case of failure to sign the Agreement and / or to furnish the requisite bank guarantee, Stage-II Connectivity shall be cancelled under intimation to the grantee without any prior notice.

vi. Two or more Applicants may apply for Stage-II Connectivity at a common bay along with an agreement duly signed between such Applicants for sharing the Dedicated Transmission Line. The Stage-II Connectivity shall be granted to such Applicants subject to availability of capacity in the Dedicated Transmission Line. In such cases, **Conn-BG1 and Conn-BG2**, as applicable as per Clause 10.8 of the Revised RE Procedure, shall be submitted by each such Applicant.

vii. The Stage-II Connectivity grantee shall furnish progress of the monitoring parameters on quarterly basis in the format given at **FORMAT-RCON-II-M** by the last day of each quarter. Failure to update progress of the monitoring parameters shall be considered as adverse progress and in such case CTU shall approach the Commission for appropriate directions. Further, the Stage-II Connectivity grantees shall be required to complete the dedicated transmission line(s) and pooling sub-station(s) as defined under para 11.2(A) of the Revised Detailed Procedure for RE. If the grantee fails to complete the dedicated transmission line within the stipulated period, the Conn-BG1 & Conn-BG2 of the grantee shall be encashed and Stage-II connectivity shall be revoked.

viii. Applicants after grant of Stage-II Connectivity to the grid shall have to furnish additional details to CTU for signing of "Connection Agreement" as per format given at **FORMAT-CON-4**. The finalized template of generation data for Solar and Wind based generation projects has been appended as additional sections at Para F (Details of Connection – Solar PV Station) and Para G (Details of Connection – Wind Generating Station) in the existing FORMAT-CON-4. The modified FORMAT CON-4 is available on our website (www.powergridindia.com >> CTU Open Access). The Applicants are advised to furnish such details as early as possible for enabling them have lead time for any type of access.

ix. The CTU will process the above information and will intimate the Connection details as per format given at **FORMAT-CON-5**. Pursuant to such Connection details, the applicant shall have to sign "Connection Agreement" with CTU prior to the physical inter-connection as per format given at **FORMAT-CON-6.** In case the connectivity is granted to the ISTS of an inter-State transmission licensee other than the CTU, a tripartite agreement shall be signed between the applicant, the Central Transmission Utility and such inter-State transmission licensee, in line with the provisions of the Regulations.

x. Applicants are required to submit the test reports supported vide undertaking as well as compliance certificate from manufacturer for all applicable provisions under the CEA (Technical Standards for Connectivity to the Grid) Regulations, 2007 (as amended) (including the provisions of LVRT/HVRT, active power injection control, dynamically varying reactive power support, limits for Harmonic & DC current injection, Flicker limits etc.) from labs accredited by Govt./NABL/other recognized agencies. In case any discrepancies / incompleteness are found in the documents / test reports submitted to CTU, the connection offer (CON-5) / connection agreement (CON-6) shall not be processed further.

xi. CEA vide order dated 09.11.2020, has mandated all power generating stations of 0.5 MW or above capacity to register themselves on CEA e-portal and get a **Unique Registration Number (URN)**. The same is required as per Regulation 11 of Technical Standards for Connectivity with the Grid Regulations, 2007 prior to physical inter-connection of the Generating station with ISTS Grid.

Annexure-I

Transmission system for LTA (226 MW) in NR

A. <u>Common Transmission system for present LTA (Appl. No. 0412100025) at Bikaner -II PS</u>

1) Augmentation of 1x500MVA (3rd) 400/220 kV ICT at Bikaner –II PS

B. <u>Common Transmission system (Part of Transmission system associated with SEZ in Rajasthan under 8.1 GW Phase-II scheme) for both LTA application</u>

1) Removal of LILO of one circuit of Bhadla-Bikaner (RVPN) 400kV D/c(Quad) line at Bikaner (PG). Extension of above LILO section from Bikaner (PG) upto Bikaner-II PS to form Bikaner-II PS – Bikaner (PG) 400kV D/c(Quad) line
2) Bikaner-II PS – Khetri 400 kV 2xD/c line (Twin HTLS* on M/c Tower)
3) 1x80MVAr switchable Line reactor on each circuit at Khetri end of Bikaner-II – Khetri 400 kV 2xD/c Line
4) Khetri- Bhiwadi 400 kV D/c line (Twin HTLS)*
5) Augmentation of 1x1500 MVA (3rd), 765/400kV ICT at Bikaner (PG)
6) Establishment of 765/400 kV, 3X1500 MVA GIS substation at Narela with 765 kV (2x330 MVAr) bus reactor and 400kV (1x125 MVAR) bus reactor.
7) Khetri – Narela 765 kV D/c line
8) LILO of 765 kV Meerut- Bhiwani S/c line at Narela
9) 1x330 MVAr Switchable line reactor for each circuit at Narela end of Khetri – Narela 765kV D/c line
10) Removal of LILO of Bawana – Mandola 400kV D/c (Quad) line at Maharani Bagh /Gopalpur S/s. Extension of above LILO section from Maharani Bagh/ Gopalpur upto Narela S/s so as to form Maharanibagh – Narela 400kV D/c(Quad) and Maharanibagh-Gopalpur-Narela 400 kV D/c (Quad) lines.
11) 2 no of line bays at Narela each for Maharanibagh – Narela 400 kV D/c (Quad) and Maharanibagh –Gopalpur**- Narela 400 kV D/c (Quad) lines formed after removal of LILO of Bawana – Mandola 400kV D/c(Quad) line at Maharani Bagh/Gopalpur S/s and Extension of above LILO section from Maharani Bagh/Gopalpur upto Narela S/s.
12) ± 300 MVAr, STATCOM along with 2x125 MVAr MSC, 1x125 MVAr MSR at Bikaner-II
13) Power reversal on ±500 KV, 2500 Balia- Bhiwadi HVDC line upto 2000 MW from Bhiwadi to Balia - Power reversal in Balia-Bhiwadi HVDC line

*with minimum capacity of 2100 MVA on each circuit at nominal voltage

**Gopalpur Substation is under implementation by DTL by LILO of Bawana – Maharanibagh 400kV D/c(Quad) at Gopalpur substation.

Additional Scheme to relieve overloading of 400 kV Bhinmal-Zerda line (Twin Moose)

1. Bypassing of 400 kV Kankroli - Bhinmal-Zerda line at Bhinmal to form 400 kV Kankroli – Zerda (direct) line #
2. Reconductoring of 400 kV Jodhpur (Surpura)(RVPN) – Kankroli S/c (twin moose) line with twin HTLS conductor*-188 km

 # with necessary arrangement for bypassing Kankroli- Zerda line at Bhinmal with suitable switching equipment inside the Bhinmal substation. with minimum capacity of 1940 MVA/ckt at nominal voltage; Upgradation of existing 400kV bay equipments each at Jodhpur (Surpura)(RVPN) and Kankroli S/s(3150 A)*